

Briefly explain two facts about polygons and polygon fill that you did not know before reading the tutorial.

First

Polygon Shapes: Polygons are divided in three shapes Convex, Concave, and Complex. The Concave are polygons are superset of Convex polygons, and Complex polygons are Concave polygons with lines that may intersect any scan line that passes through the polygon. This relationship identifies that, if we construct an algorithm for complex polygons then this algorithm is said to work for Concave and Convex polygons.

Second:

Odd parity algorithm: This algorithm test entails by drawing a line segment from any point that lies outside the polygon to a point that we wish to determine whether it is inside the polygon or not. By keeping the parity even and changing to odd every time edges are found, then this would allows us to determine when we are inside and out of a polygon.

Conclusion:

Since the odd parity algorithm is said to work for Complex polygons then, this is said to work for all other polygon shapes. This is a very efficient and fast algorithm that would help us to fill in polygon shapes in a recursive and simplified optimum way.